SanTam Net Alert

AN INTELLIGENT EMAIL TO SMS NOTIFICATION SYSTEM

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BONAFIDE CERTIFICATE

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ABSTRACT

SanTam NetAlert is a full-stack intelligent notification system designed to bridge the communication gap between email-based alerts and real-time mobile SMS notifications. In today's fast-paced academic and professional environment, critical email notifications often go unnoticed due to poor internet connectivity, device silent mode, or email overload. This project addresses this challenge by continuously monitoring a configured Gmail inbox using the IMAP protocol and automatically forwarding important emails as SMS messages to a mobile phone — even without internet access on the receiving device.

The system employs a keyword-based intelligent filtering engine that scans incoming email subjects and body content for user-defined trigger words such as URGENT, Assignment, Deadline, Exam, and Result. Only emails matching these configured keywords trigger an SMS alert, ensuring that users receive meaningful notifications without noise or spam. SMS delivery is achieved through the httpSMS open-source Android gateway, which uses the user's own Android phone as an SMS transmitter via its SIM card, completely eliminating the need for expensive third-party SMS services.

The backend of the system is built using Node.js with Express.js, providing a RESTful API architecture with endpoints for email management, SMS triggering, delivery log retrieval, and dynamic settings configuration. The frontend dashboard is developed using React.js with Vite, offering real-time statistics including total emails received, SMS sent count, failed SMS count, and a complete SMS delivery log with SUCCESS and FAILED status indicators. The environment-based configuration system using a .env file ensures secure management of sensitive credentials including Gmail App Passwords and SMS API keys.

This project demonstrates the practical integration of networking protocols (IMAP, HTTP, REST), web technologies, and mobile communication to create a cost-effective, reliable real-time notification system applicable to students, IT professionals, and business users requiring offline SMS alerts for time-sensitive emails.

TABLE OF CONTENTS

S. No	PARTICULARS	PAGE NO
	ABSTRACT	v
	TABLE OF CONTENT	vi
	LIST OF TABLES	viii
	LIST OF FIGURES	ix
1	**INTRODUCTION**	1
1.1	Introduction to the Project	2
1.2	Objective of the Project	2
1.3	Scope of the Project	3
1.4	Need for Automation	3
2	**LITERATURE REVIEW**	4
2.1	Existing Manual Analysis	5
2.2	Limitation of Traditional Tools	5
2.3	Need for Automatic Detection	6
2.4	Advantage of Proposed Approach	6
3	**PROPOSED SYSTEM**	7
3.1	Overview of SanTam NetAlert	8
3.2	System Architecture	8
3.3	Core Features of the System	9
3.4	Advantages of the Proposed System	10
3.5	Applications of the System	11
3.6	System Diagrams	12
4	**METHODOLOGY**	13
4.1	System Workflow Description	14
4.2	Step-by-Step Implementation	15
4.3	Code Explanation	15
4.4	Tools and Technologies Used	17
5	**MODULE DESCRIPTION**	19
5.1	Email Monitoring Module	20
5.2	SMS Gateway Module	20
5.3	Keyword Filter Module	21
5.4	Dashboard and Settings Module	22
6	**RESULT AND DISCUSSION**	25

S. No	PARTICULARS	PAGE NO
6.1	System Performance Analysis	26
6.2	Discussion of Implementation	27
6.3	Advantages of the Developed System	28
7	**CONCLUSION**	29
7.1	Conclusion of the Project	30
7.2	Key Achievements	30
7.3	Limitations of the Current System	31
8	**FUTURE ENHANCEMENT**	34
8.1	Future Scope	35
	APPENDIX	37
	REFERENCES	51

LIST OF TABLES

Table No	Table Name	Page No
4.4.1	Technologies Used in SanTam NetAlert	17
4.4.2	API Endpoints Reference	18
6.1.1	System Performance Metrics	26

LIST OF FIGURES

Figure No	Figure Name	Page No
3.2.1	System Architecture of SanTam NetAlert	9
3.6.1	Data Flow Diagram of SanTam NetAlert	12
4.1.1	Workflow of SanTam NetAlert System	14
5.1.1	Email Monitoring Module Workflow	21
A.1	System Flowchart of SanTam NetAlert	38
A.2	Email Processing Flowchart	38
A.3	SMS Sending Flowchart	38
A.4	Entity Relationship Diagram	39
A.5	Data Flow Diagram (Level 0)	39
A.6	Data Flow Diagram (Level 1)	40
A.7	Dashboard Interface	40
A.8	SMS Delivery Logs Interface	41
A.9	Settings Configuration Page	41